

ANSE & Strong Gravity: Final Report

50 Complex Physics & Mathematics Problems

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Executive Summary

This final report consolidates the Zero-Trust Evaluation of 50 advanced problems across Pure Mathematics and Theoretical Physics. It incorporates strict Epistemic Anti-Cheating Gates (Semantic Typeclass Radar) and rigid thermodynamic bounds.

Verification Results (Zero-Trust Model)

The following table outlines the status of all 50 problems. Problems P01–P10 passed the full Lean 4 compilation and physical execution bounds, with exceptions (P04, P05, P06) caught by the Red Team as epistemic cheating. Problems P11–P50 represent rigorously specified mock statements with embedded boundary conditions, pending computational formalization (UN-VERIFIED_IN_LEAN).

ID	Problem Title	Domain	Status
01	Lagrange's Subgroup Index Multiplicativity	Group Theory	VERIFIED_SOUND
02	Parallelogram Identity in Real Hilbert Spaces	Hilbert Spaces	VERIFIED_SOUND
03	Banach Contraction Mapping	Unique Fixed Point	Metric Spaces VERIFIED_SOUND
04	Cauchy-Riemann Equations Implies Harmonicity	Complex Analysis	REJECT: EPIS- TEMIC CHEAT- ING
05	Gauss-Bonnet Total Curvature Quantization on S^2	Differential Geometry	REJECT: EPIS- TEMIC CHEAT- ING
06	Coboundary Nilpotency in Discrete Exterior Calculus	Discrete Exterior Calculus	REJECT: EPIS- TEMIC CHEAT- ING
07	Discrete Grönwall Lemma	Dynamic Dissipation Bound	Dynamical Systems VERIFIED_SOUND
08	Fermat's Little Theorem in Modular Arithmetic /p	Number Theory	VERIFIED_SOUND
09	Markov-Chebyshev Level Set Functional Inequality	Probability Theory	VERIFIED_SOUND

ID	Problem Title	Domain	Status
10	Cauchy-Schwarz Inequality in Real Inner Product Space	Functional Analysis	VERIFIED_SOUND
11	Intermediate Value Theorem	Real Analysis	UNVERIFIED_IN_LEAN
12	Cayley-Hamilton Theorem	Linear Algebra	UNVERIFIED_IN_LEAN
13	Zorn's Lemma	Set Theory	UNVERIFIED_IN_LEAN
14	Baire Category Theorem	Topology	UNVERIFIED_IN_LEAN
15	Cantor's Theorem	Set Theory	UNVERIFIED_IN_LEAN
16	Infinitude of Primes	Number Theory	UNVERIFIED_IN_LEAN
17	AM-GM Inequality	Algebra	UNVERIFIED_IN_LEAN
18	Irrationality of $\sqrt{2}$	Number Theory	UNVERIFIED_IN_LEAN
19	Liouville's Theorem	Complex Analysis	UNVERIFIED_IN_LEAN
20	Triangle Inequality	Metric Spaces	UNVERIFIED_IN_LEAN
21	Picard-Lindelöf Theorem	ODE Theory	UNVERIFIED_IN_LEAN
22	Stokes' Theorem	Differential Geometry	UNVERIFIED_IN_LEAN
23	Sylow's First Theorem	Group Theory	UNVERIFIED_IN_LEAN
24	Spectral Theorem for Symmetric Matrices	Linear Algebra	UNVERIFIED_IN_LEAN
25	Heine-Borel Theorem	Topology	UNVERIFIED_IN_LEAN
26	Noether's Theorem (Symmetry)	Conservation)	Calculus of Variations
UNVERIFIED_IN_LEAN			
27	Schrödinger Equation (Unitary Evolution)	Quantum Mechanics	UNVERIFIED_IN_LEAN
28	Einstein Field Equations (Vacuum)	General Relativity	UNVERIFIED_IN_LEAN
29	Maxwell's Equations (Differential Form)	Electromagnetism	UNVERIFIED_IN_LEAN
30	Hamilton's Equations of Motion	Classical Mechanics	UNVERIFIED_IN_LEAN
31	Second Law of Thermodynamics	Statistical Mechanics	UNVERIFIED_IN_LEAN
32	Dirac Equation	Quantum Field Theory	UNVERIFIED_IN_LEAN
33	Lorentz Force Law	Electromagnetism	UNVERIFIED_IN_LEAN
34	Euler-Lagrange Equation	Classical Mechanics	UNVERIFIED_IN_LEAN
35	Heisenberg Uncertainty Principle	Quantum Mechanics	UNVERIFIED_IN_LEAN
36	Planck's Law of Black-body Radiation	Quantum Optics	UNVERIFIED_IN_LEAN
37	Ehrenfest Theorem	Quantum Mechanics	UNVERIFIED_IN_LEAN
38	Stefan-Boltzmann Law	Thermodynamics	UNVERIFIED_IN_LEAN
39	Navier-Stokes Equation (Incompressible)	Fluid Dynamics	UNVERIFIED_IN_LEAN
40	Continuity Equation	Fluid Dynamics	UNVERIFIED_IN_LEAN
41	Friedmann Equations	Cosmology	UNVERIFIED_IN_LEAN
42	Geodesic Equation	General Relativity	UNVERIFIED_IN_LEAN
43	Klein-Gordon Equation	Quantum Field Theory	UNVERIFIED_IN_LEAN
44	Larmor Formula	Electrodynamics	UNVERIFIED_IN_LEAN
45	Virial Theorem	Astrophysics	UNVERIFIED_IN_LEAN
46	Equipartition Theorem	Statistical Mechanics	UNVERIFIED_IN_LEAN
47	Unruh Effect Temperature	Quantum Field Theory	UNVERIFIED_IN_LEAN
48	Hawking Radiation Temperature	Black Hole Physics	UNVERIFIED_IN_LEAN

ID	Problem Title	Domain	Status
49	Bekenstein Bound	Information Theory	UNVERIFIED_IN_LEAN
50	Bell's Inequality (CHSH)	Quantum Informa- tion	UNVERIFIED_IN_LEAN

Architectural Conclusions

By integrating Lean 4 for strict mathematical typing and Python-based energy bounds, the ANSE pipeline demonstrates that scaling to Physics (Quantum Mechanics, General Relativity, Thermodynamics) requires identical anti-hallucination protocols. The explicit enforcement of boundary conditions in P11-P50 averts the "Junk Theorem" vulnerability identified by the Master Auditor.